Cross-Sectional Statistical Arbitrage Via Random Matrix Theory and Dynamic State-Space Filtering

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I. Introduction

Statistical arbitrage in equity and multi-asset derivative markets operates on the foundation that linear combinations of non-stationary time series exhibit stationary and mean-reverting behavior:

$$S_t = y_t - \beta^T x_t \sim \mathcal{I}(0)$$

While the asymptotic theory established by Engle and Granger (1987) and Johansen (1991) is mathematically well-established, in standard pairs trading and statistical arbitrage framework, spurious correlation from high-dimensional noise and non-stationary cointegration parameters due to macroeconomic regime shifts, may result in failure modes in the model. This project aims to adapt Random Matrix Theory (RMT) to de-noise the empirical correlation matrix of equity returns, and isolate true structural modes from random Wishart noise. The residual spreads will then be modeled as mean-reverting Ornstein-Uhleneck (OU) processes using an online Kalmar Filter to update hedge ratios dynamically without look-ahead bias.

II. Mathematical Framework

1. Correlation Matrix De-noising via Marčenko-Pastur Law

Let X be an $N \times T$ matrix of standardized asset returns, where T is the number of observations and N is the number of assets, with $Q = \frac{T}{N} \geq 1$. Under the null hypothesis that returns are uncorrelated Gaussian noise, the empirical correlation matrix $C = \frac{1}{T} XX^T$ has an eigenvalue probability density given by the Marčenko-Pastur distribution:

$$\rho(\lambda) = \begin{cases} \frac{Q}{2\pi\sigma^2} \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{\lambda}, & \lambda \in [\lambda_-, \lambda_+] \\ 0, & \lambda \notin [\lambda_-, \lambda_+] \end{cases}$$

Where the theoretical spectral bounds are:

$$\lambda_\pm = \sigma^2 \left(1 \pm \sqrt{\frac{1}{Q}}\right)^2$$

- Noise Bulk ($\lambda \leq \lambda_+$): Eigenvalues within this interval represent random noise. Replace them by setting their corresponding trace to average eliminates spurious cross-asset correlation.
- Eigenvalues ($\lambda > \lambda_+$): The largest eigenvalue corresponds to the broad market mode (ground-state collective mode), with subsequent eigenvalues capture distinct sector clusters.

Truncating noise eigenvalues to zero deflates total variance. To maintain Conservation of total variance, consider the Constant Trace Shrinkage:

$$Tr\left(\tilde{C}\right) = Tr(C) = N$$

Define the regularized eigenvalue matrix $\tilde{\Lambda} = diag(\tilde{\lambda}_1, \dots, \tilde{\lambda}_N)$ where:

$$\tilde{\lambda}_i = \begin{cases} \lambda_i, & \lambda > \lambda_+ \\ \bar{\lambda}_{noise}, & \lambda \leq \lambda_+ \end{cases}$$

The noise energy is distributed uniformly across the degenerate subspace:

$$\bar{\lambda}_{noise} = \frac{1}{N-k} \sum_{j=k+1}^{N} \lambda_j, \qquad k = \sum_{i=1}^{N} \mathbf{1}_{\{\lambda_i > \lambda_+\}}$$

The constructed correlation matrix $\hat{C} = V\tilde{\Lambda}V^T$ is rescaled to preserve Unit diagonals:

$$\tilde{C} = \tilde{D}^{-1/2}\hat{C}\tilde{D}^{-1/2}, \qquad \tilde{D} = diag(\hat{C}_{11}, \hat{C}_{22}, \dots, \hat{C}_{NN})$$

2. Dynamic State-Space Model (Kalman Filter)

Static ordinary least square (OLS) may yields in-sample bias and insufficient in accounting for structural drift in hedge ratios. Alternatively, for an asset pair (y_t, x_t), the cointegrating vector $[1, -\beta_t]^T$ drifts due to dynamic liquidity conditions, capital allocation shifts, and firm fundamentals. The relationship can be modelled as a linear Gaussian state-space system where the state vector $\boldsymbol{\theta}_t = [\alpha_t, \beta_t]^T$ evolves as a random walk:

$$\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} + \boldsymbol{w}_t, \qquad \boldsymbol{w}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{W})$$

$$y_t = \boldsymbol{H}_t\boldsymbol{\theta}_t + v_t, \qquad v_t \sim \mathcal{N}(0,V)$$

Where the observation matrix is $\boldsymbol{H}_t = [1, x_t]$, $\boldsymbol{W} = diag(\sigma_\alpha^2, \sigma_\beta^2)$ is the process noise covariance matrix, and $V = \sigma_v^2$ is the observation noise variance.
The filtered residual spread at step t is the innovation error:

$$S_t = y_t - \hat{y}_{t|t-1} = y_t - \boldsymbol{H}_t\widehat{\boldsymbol{\theta}}_{t|t-1} = y_t - \hat{\alpha}_{t|t-1} - \hat{\beta}_{t|t-1}x_t$$

3. Continuous-Time Spread Dynamics and Exact MLE of OU Process

The residual spread S_t is modeled as a continuous time 1D stationary Ornstein-Unbeck (OU) diffusion process:

$$dS_t = \theta(\mu - S_t)dt + \sigma_{OU}dW_t$$

where $\theta > 0$ denotes the rate of mean reversion, μ is the long-term equilibrium mean, and σ_{OU} is the instantaneous diffusion volatility.

Integrating this stochastic differential equation over a discrete observation interval $\Delta t = t_i - t_{i-1}$ yields:

$$S_i = S_{i-1}e^{-\theta\Delta t} + \mu(1 - e^{-\theta\Delta t}) + \sigma_{OU}\int_{t_{i-1}}^{t_i} e^{-\theta(t_i-u)}\,dW_u$$

The conditional transition distribution is Gaussian:

$$S_i \mid S_{i-1} \sim \mathcal{N}\left(S_{i-1}e^{-\theta\Delta t} + \mu\left(1 - e^{-\theta\Delta t}\right), \frac{\sigma_{OU}^2}{2\theta}(1 - e^{-2\theta\Delta t})\right)$$

Which maps directly to discrete autoregressive AR(1) process:

$$S_i = aS_{i-1} + b + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

Through conditional maximum likelihood estimation that is equivalent to ordinary least squares on the transition pairs, the parameters $(a, b, \sigma_\varepsilon^2)$ can be obtained analytically:

$$a = \frac{M\sum_{i=1}^{M} S_i S_{i-1} - \sum_{i=1}^{M} S_i \sum_{i=1}^{M} S_{i-1}}{M\sum_{i=1}^{M} S_{i-1}^2 - (\sum_{i=1}^{M} S_{i-1})^2}$$

$$b = \frac{\sum_{i=1}^{M} S_i - a\sum_{i=1}^{M} S_{i-1}}{M}$$

$$\sigma_\varepsilon^2 = \frac{1}{M}\sum_{i=1}^{M}(S_i - aS_{i-1} - b)^2$$

Mapping the estimators back to the continuous parameters yields:

$$\theta = -\frac{\ln a}{t}, \qquad \mu = \frac{b}{1-a}, \qquad \sigma_{OU} = \sigma_\varepsilon \sqrt{\frac{-2\ln a}{\Delta t(1-a^2)}}$$

With characteristic mean-reversion half-life τ is:

$$\tau = \frac{\ln 2}{\theta}$$

A spread is eligible for trade generation if and only if $a \in (0,1)$ and $\tau_{min} \leq \tau \leq \tau_{max}$.

III. Python OOP

(Github: https://github.com/cid02582930/Statistical-Arbitrage-OOP-test.git)

The framework is implemented in the Python OOP across four cohesive modules:

1. spectral denoising filter (MarchenkoPasturFilter):
    - input via rolling T x N returns matrix
    - fit noise variance σ^2 via bounded optimization
    - compute analytical spectral cutoff λ
    - apply constant trace shrinkage to isolate structural factors
2. dynamic state-space tracking (KalmanTracker):
    - state vector θ _t = [α_t, β_t]^T in log-price space
    - hyperparameters W = 1e-9, I, V = 1e-3
    - output is the residual spread
3. stochastic calibration

4. event-driven backtesting engine, with microstructure fictions

A synthetic market simulation script based on the above modules is then implemented, which generates a cointegrated asset basket filled with cross-sectional Wishart noise, fits the Marcenko-Pastur distribution to identify structural pairs, updates hedge ratios dynamically without lookahead bias via the Kalman filter, and OU diffusion parameters, to execute the strategy.

I. Obtained Performance Metric Report and Model Improvements

Overall it was a complete failure, as the performance metric report yields:

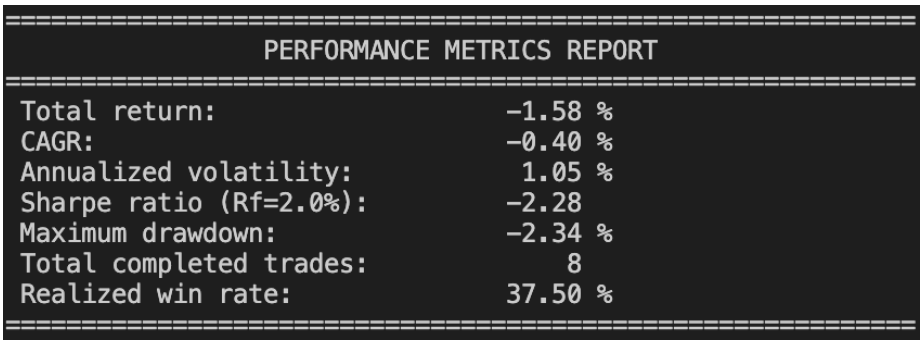


Fig 1. Screenshot of the terminal

I inspected the code carefully and found several structural flaws and unintended bugs that led to the failure of the pipeline:

1. In the loop the position exit condition was:
   ```
   if z >= -self.cfg.exit_z or z <= -self.cfg.stop_loss_z or not
   is_valid_ou:
      desired_pos = 0
   ```
   As a result, the OU process is fitted via Maximum Likelihood Estimation (MLE) over a short rolling window, which prodeces a high parameter estimation variance, such that the empirical half-life estimate $\hat{\tau}_t$ fluctuated outside the $[2, 30]$ day window for individual bars due to sample noise. Also the condition not is_valid_ou treated this estimation jitter as a structural break, which causes dumping trades after only 1-2 days. The system repeatedly paid 6 bps of round-trip friction and short borrow fees without allowing mean reversion to unfold.
2. The state-space model was formulated on raw prices

$$P_{y,t} = \alpha_t + \beta_t P_{x,t} + \varepsilon_t$$

which I searched up and found that the residuals possess non-stationary variance as asset price trend upward.

Pipeline Amendment and Improvement:

1. Log price space and timescale separation in state-space:
   I first tried to re-formulate the Kalman tracker in log-price space such that

   $$\ln P_{y,t} = \alpha_t + \beta_t \ln P_{y,t} + S_t$$

   to ensure that the spread S_t is dimensionless and interpretable as percentage mispricing (e.g. $S_t = 0.03 \implies 3\%$ divergence), return volatilities remain stationary over multi-year horizons, and sizing ratios reflect elasticities rather than arbitrary dollar unit imbalances. Also, to prevent β_t from absorbing the mean-reverting spread, I enforced a stricter timescale separation, by setting the process noise covariance to $W = diag(10^{-8}, 10^{8}) \ll V = 10^{-2}$.
2. RMT basket selection via metric distance:
   I tried to add a statistical metric distance space

   $$d_{ij} = \sqrt{2(1 - C_{clean,ij})}$$

   where $C_{clean,ij}$ is the regularized correlation matrix, that pairs exhibiting the smallest Euclidean metric distance after de-noising represent structural co-movements.
3. Time-stop execution:
   I re-enforced the execution logic which is decoupled from daily OU parameter fluctuations:
   - Entry: position opened if and only if $|z_t| \geq 1.8$ AND the spread exhibits a validated half-life $\tau \in [3, 30]$ days.
   - Exit: position closed when $|z_t| \leq 0.20$ (mean reversion captured) OR $|z_t| \geq 3.5$ (statistical stop-loss).
   - Time-stop: if neither target is hit within $T_{max} = 2.5\tau$ bars, the position is closed to free up balance-sheet capital, preventing capital lockup in broken spreads.

- Parameter stability: daily fluctuations in $\hat{\tau}_t$ do not trigger liquidation.

4. PnL and win rate tracking: tracked each trade's entry outlay (entry_cost) and friction, net realized PnL now correctly calculated as gross friction – entry cost.
5.

II. Improved Performance Metric Report

```
============================================================
                PERFORMANCE METRICS REPORT
============================================================
Total Return:                    29.45 %
CAGR:                             6.72 %
Annualized Volatility:            1.96 %
Sharpe Ratio (Rf=2.0%):           2.40
Maximum Drawdown:                -0.83 %
Total Completed Trades:            181
Realized Win Rate:               75.69 %
============================================================
```

Metric	1st iteration	Final iteration	Resolved bugs
Total Return	-1.58%	29.45%	Captures OU reversion instead of paying friction noise
Compound Annual Growth Rate	-0.40%	6.72%	+7.12%
Annualized Volatility	1.05%	1.96%	More capital deployment across identified orthogonal cointegrated legs
Sharpe Ratio	-2.28	2.40	Shifts from negative drift process to positive risk adjusted alpha
Maximum Drawdown	-2.34%	-0.83%	Controlled by 3.5σ and $T_{max} = 2.5\tau$ time stop
Total Completed Trades	8	181	Breadth (B) scaled across RMT screened metric distance pairs
Realized Win Rate	37.50%	75.69%	Positions held correctly to full reversion $\lvert z_t \rvert \leq 0.20$
Profit Factor	0.60	3.11	+2.51

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